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U–Pb zircon ages of Kangâmiut dykes and detrital zircons in metasediments in the Palaeoproterozoic Nagssugtoqidian Orogen (West Greenland) Clues to the pre-collisional history of the orogen[1]

Allen P. Nutman [a,*], Feiko Kalsbeek [b,c], Mogens Marker [c], Jeroen A.M. van Gool [c], David Bridgwater [d]

[a] *Research School of Earth Sciences, Australian National University, Canberra, A.C.T. 0200, Australia*
[b] *Geological Survey of Denmark and Greenland, Thoravej 8, 2400 Copenhagen, NV, Denmark*
[c] *Danish Lithosphere Centre, Øster Voldgade 10, 1350 Copenhagen, K, Denmark*
[d] *Geological Museum, Øster Voldgade 5–7, 1350 Copenhagen, K, Denmark*

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Abstract

In the southern marginal zone of the Palaeoproterozoic Nagssugtoqidian orogen and its foreland (West Greenland), Archaean gneisses were intruded by the syn-kinematic Kangâmiut dyke swarm. Leucodioritic cores of two dykes have yielded U–Pb zircon ages of 2036 ± 5 Ma and 2046 ± 8 Ma (2σ). Dyke emplacement and associated crustal movements thus occurred well before the main plutonic and tectonic activity within the orogen ($\sim$1900–1800 Ma).

In the central Nagssugtoqidian orogen, pelitic to semi-pelitic metasedimentary rocks were intruded by 1920–1900 Ma quartz diorites and tonalites of the Arfersiorfik association, which have chemical and isotopic compositions comparable with igneous suites formed in arc environments. The supracrustal and intrusive rocks are separated from Archaean gneisses by décollements of high-grade mylonite and annealed marble, and are allochthonous units. Detrital zircons in two samples of metasedimentary rocks from these allochthons are mostly large, poorly rounded, 2100–1950 Ma old with only a few Archaean grains. Deposition of the metasediments must have taken place between $\sim$1950 and $\sim$1920 Ma, before the intrusion of rocks of the Arfersiorfik association. The zircons may have been shed from presently unrecognised pre-Arfersiorfik association igneous complexes (arcs?), whose emplacement coincided with intrusion of the $\sim$2040 Ma Kangâmiut dykes into deforming Archaean crust in what later became the southern margin of the orogen.

Quartz-rich psammitic metasedimentary rocks are also present in the central Nagssugtoqidian orogen. Detrital zircons from one sample yielded a few 2150–2250 Ma grains, but are predominantly Archaean. They include a high proportion of 3300–3400 Ma grains (the oldest material yet found in the orogen). Thrust-bounded panels of metasedimentary rocks in Archaean gneisses occur at the boundary between the southern and central parts of the orogen coinciding with the northern limit of the Kangâmiut dykes. Of the detrital zircons in one metasediment sample from these panels a few yielded ages around 2100–2000 Ma, some were Archaean, and $>50\%$ were $\sim$2400 Ma.

* Corresponding author. Fax: +61 2 62490738;
e-mail: allen.nutman@anu.edu.au

[1] The death of Professor David Bridgwater occurred in 1998.

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Deposition of these two sediment suites took place after ~2100 Ma. The detrital zircon population shows that the sources for these sediments were different from that of the above-mentioned allochthonous rocks. Neither the 3300–3400 Ma nor the ~2400 Ma detrital grains can be matched with any rocks in Greenland, indicating a distal source for these sediments. © 1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

The Palaeoproterozoic Nagssugtoqidian orogen of West Greenland (Fig. 1) is a ~200 km wide belt north of the Archaean craton of southern Greenland (Ramberg, 1949; Escher et al., 1976a; Korstgård, 1979a). The orogen is characterised by a strong overall east–northeast trending structural grain, and it is now generally accepted that it continues eastwards under the Greenland Inland Ice and links up with the Nagssugtoqidian (or Ammassalik) belt in East Greenland. This Greenland belt, and ones of the same general age and character in northern Europe and arctic Canada, may belong to a once continuous, extensive Palaeoproterozoic curvilinear orogenic belt in the North Atlantic region (e.g. Bridgwater et al., 1990). The orogen consists predominantly of Archaean gneisses that were reworked under amphibolite to granulite facies conditions during Nagssugtoqidian orogeny. The southern foreland of the Nagssugtoqidian orogen consists of Archaean gneisses intruded by a suite of mafic dykes, the Kangâmiut dykes, that are progressively reworked into the orogen. Within the central Nagssugtoqidian orogen there are allochthonous juvenile Palaeoproterozoic arc rocks and associated metasedimentary rocks (Kalsbeek et al., 1987; Kalsbeek and Nutman, 1996; van Gool et al., 1996; Marker et al., 1997a; Whitehouse et al., in press). To the north of the Nagssugtoqidian orogen lies the Palaeoproterozoic Rinkian mobile belt (Fig. 1), however, the relationship between these orogens remains enigmatic (Grocott and Davies, in press).

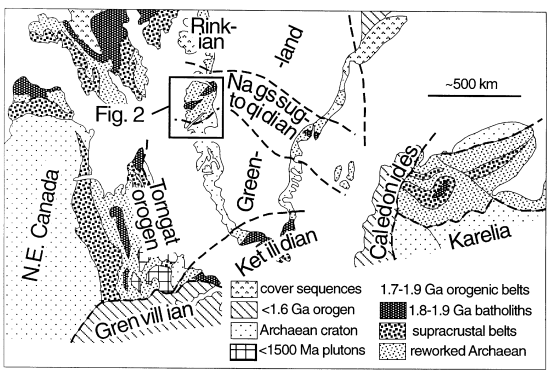


Fig. 1. Sketch map showing the location of the Nagssugtoqidian orogen in the North Atlantic region, adapted from Mengel (1996).

The Nagssugtoqidian orogen is interpreted to be a collisional orogenic belt in which elements associated with a period of subduction and continental collision can be recognised (Bridgwater et al., 1973; Kalsbeek et al., 1987; van Gool et al., this issue). The timing of subduction is constrained by the age of the two arc-related igneous suites, the Arfersiorfik association and the Sisimiut charnockite suite, which intruded at ~1920–1900 Ma (Kalsbeek and Nutman, 1996; Whitehouse et al., in press). A period of thrusting and large-scale folding at peak metamorphic conditions, eventually resulting in the present structural grain, is interpreted to represent a phase of continental collision. The peak of metamorphism has been dated at ~1850–1800 Ma (Taylor and Kalsbeek, 1990; Kalsbeek and Nutman, 1996). A latest phase of magmatism is represented by granitic pegmatites which occur throughout the orogen and are dated in several locations at ~1780–1760 Ma (Kalsbeek and Nutman, 1996; Whitehouse et al., in press).

The period between the end of the Archaean and the onset of Nagssugtoqidian orogeny (ca. 2500 Ma to 1920 Ma) appears to be marked by little tectonic activity in the area now covered by the orogen. This paper investigates the tectonic activity immediately prior to the Nagssugtoqidian

orogeny by means of SHRIMP U–Pb zircon dating on samples of Kangâmiut dykes and metasedimentary rocks from several suites in the centre of the orogen.

2. The Nagssugtoqidian orogen

2.1. Southern Nagssugtoqidian orogen

Following Ramberg (1949), the amphibolite to granulite facies Palaeoproterozoic Nagssugtoqidian orogen was divided by Marker et al. (1995) into a southern, a central and a northern segment (respectively SNO, CNO and NNO; Fig. 2). The southern Nagssugtoqidian orogen consists mainly of Archaean gneisses, for which ages of ~2700 to 2850 Ma are most common, comparable with the gneisses in the southern foreland (Kalsbeek and Nutman, 1996; Connelly and Mengel, 1996). It contains variably deformed remnants of the Kangâmiut dykes that form a swarm in the northern part of the Archaean craton (Ramberg, 1949; Escher et al., 1975). The northern margin of the SNO is a south-vergent thrust belt, referred to in the west as the Ikertôq thrust zone by Grocott (1977), where metasedimentary rocks occur in thin panels (≤100 m) interleaved with Archaean gneisses (Marker et al., 1997a,b). To the south, within the SNO, Kangâmiut dykes are widespread, whereas to the north (CNO) the presence of the dykes has not been demonstrated (Grocott, 1977; Marker et al., 1995; van Gool et al., 1996).

2.2. Central Nagssugtoqidian orogen

The central Nagssugtoqidian orogen is dominated by Archaean gneisses of ages similar to those in the SNO, but also contains significant volumes of Palaeoproterozoic ortho- and paragneisses (Kalsbeek et al., 1987; Marker et al., 1995; Kalsbeek and Nutman, 1996; van Gool et al., 1996). In the eastern Nordre Strømfjord–Arfersiorfik area (Fig. 2), a suite of 1920–1900 Ma tonalitic to dioritic gneisses have been named the Arfersiorfik association (Kalsbeek and Nutman, 1996). These rocks show geochemical and isotope compositions similar to arc rocks (Kalsbeek et al., 1987) and are intrusive into a sequence of supracrustal rocks with which they are tectonically interlayered.

The Arfersiorfik association and the spatially associated metasedimentary rocks form a separate lithotectonic unit, referred to as the Ussuit unit by Manatschal et al. (in press) and van Gool et al. (this issue). The rocks are separated from structurally underlying Archaean gneisses (U–Pb zircon ages predominantly between ~2900 and 2700 Ma, Kalsbeek et al., 1987; Kalsbeek and Nutman, 1996; Connelly and Mengel, 1996) by a décollement of high grade mylonite and annealed marble that has been interpreted as a major thrust (Kalsbeek and Nutman, 1996; Manatschal et al., in press; Passchier et al., in press; van Gool et al., this issue). This décollement and the tectonostratigraphic sequence of Archaean and Palaeoproterozoic gneisses was subsequently repeated by thrust imbrication (van Gool et al., this issue), followed by non-cylindrical tight to isoclinal folding and later open warping, giving rise to a complex basin and dome style fold interference pattern. On the basis of this tectonic relationship and the geochemical and isotopic data, Kalsbeek and Nutman (1996) proposed that the Ussuit unit represents an allochthon containing juvenile Palaeoproterozoic arc rocks emplaced during or before the 1850–1800 Ma peak metamorphism.

In the southwestern CNO, around the town of Sisimiut, the granulite facies Sisimiut charnockite suite (Fig. 2), dominated by tonalitic to dioritic phases, has given SHRIMP U–Pb zircon ages of 1920–1900 Ma, indistinguishable in age from the Arfersiorfik association (Kalsbeek and Nutman, 1996; Whitehouse et al., in press). The two units are also comparable in chemical and isotopic composition.

2.3. Northern Nagssugtoqidian orogen

The NNO consists also mainly of Archaean orthogneisses for which ages of ~2700 to 2850 Ma have been obtained (Kalsbeek et al., 1987; Kalsbeek and Nutman, 1996; Connelly and Mengel, 1996), and which are interleaved with minor supracrustal sequences of unknown age. No

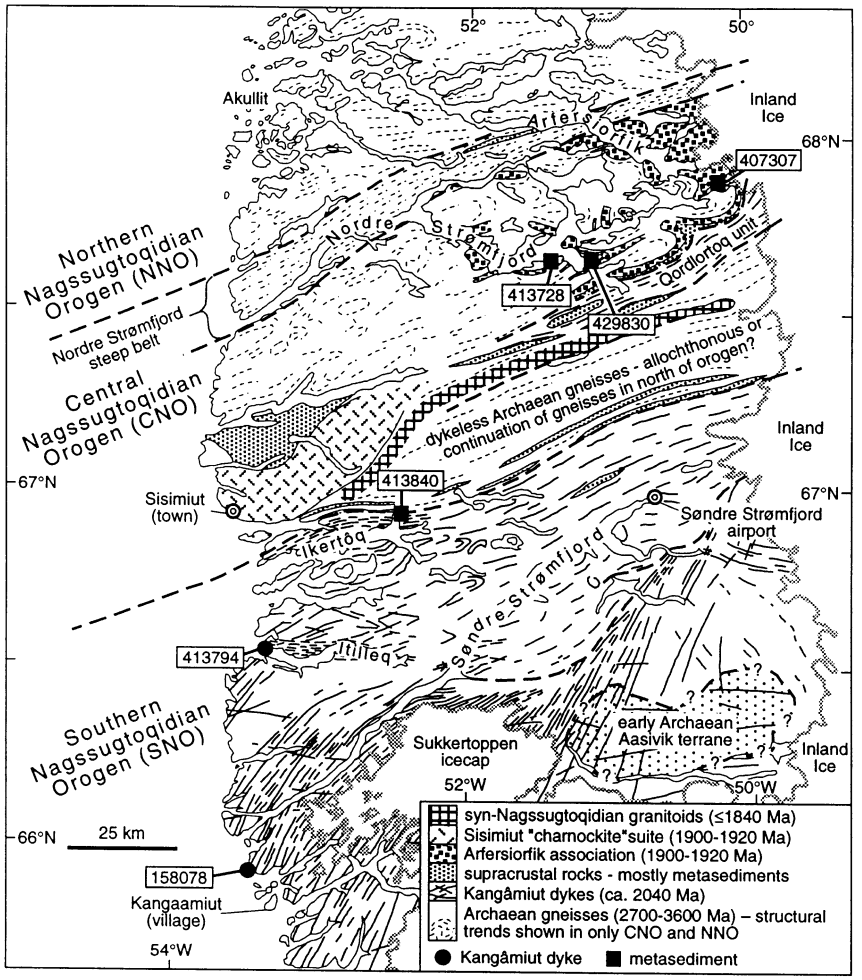


Fig. 2. Map of the Nagssugtoqidian orogen, West Greenland. Compiled from material in Escher (1971), Allaart (1982), van Gool et al. (1996) and Rosing et al. (in press).

Palaeoproterozoic rocks have yet been recognised. The Archaean orthogneisses are probably continuous with those of similar age and composition that are tectonically interleaved with the Arfersiorfik association in the northeastern CNO.

3. Pre-Nagssugtoqidian lithotectonic units

The time between the end of the Archaean and the start of Nagssugtoqidian orogeny was tectonically a relatively quiet period in the area occupied by the orogen. Lithological units formed in this period are important for two reasons: first as indicators of activity prior to the Nagssugtoqidian orogeny, and second because they can act as time markers to distinguish between events related to the Nagssugtoqidian orogeny and older tectonothermal activity.

3.1. Kangâmiut dykes

The Palaeoproterozoic Kangâmiut dyke swarm cuts the Archaean gneisses (Fig. 2) in a 150 km

wide belt south of the CNO–SNO boundary. These dykes have proven to be an excellent time and strain marker (Ramberg, 1949; Escher et al., 1976b). Northwards, within the SNO, the dykes become progressively more deformed and recrystallised under amphibolite and then locally granulite facies conditions (Korstgård, 1979b). Individual dykes are generally <50 m thick, but a few are 100 m wide. Some of the >10 m wide dykes consist of hornblende–augite dolerite margins and contain heterogeneous cores of hornblendite/gabbro and leucodiorite. Primary igneous hornblende suggests that some were hydrous at the time of intrusion. Geochemical and whole rock isotope studies (e.g. Kalsbeek et al., 1978; Bridgwater et al., 1995) show that the margins and centres of the dykes are consanguineous, and there is no evidence that the cores were derived from remelting of the local Archaean gneisses. Kalsbeek et al. (1978) reported a Rb–Sr age of 1950 ± 60 Ma for two dykes near Kangaamiut village.

The dioritic cores commonly develop strong shear fabrics indicating both dextral and sinistral movements at the time they crystallised. Where dykes of the same swarm intersect in the foreland, foliated young dykes cut the foliation in old dykes, which demonstrates that these foliations and their mineral assemblages developed during emplacement of the dykes, and not during a later unrelated tectonic event (Windley, 1970). Thus the dykes are syn-tectonic (cf. Escher et al., 1976b), in agreement with Hanmer et al. (1997), who, based on the geometry of the dyke arrays, suggested that they intruded in a sinistral transpressive environment. Crustal extension associated with the emplacement of these dykes has been estimated as varying from 2–3% to up to 15% in some areas (Allaart and Jensen, 1979; Escher et al., 1975).

3.2. Metasedimentary rocks

Metasedimentary rocks are found in a number of supracrustal suites, which have been strongly tectonised and fragmented during Nagssugtoqidian deformation. The suites can be related to different tectonic settings, most of which coincide with major tectonic boundaries (van Gool et al., 1996; Marker et al., 1997b). The largest volumes of supracrustal rocks were identified in the CNO and the northern part of the SNO (Marker et al., 1997a).

The supracrustal suite in the Nordre Strømfjord–Arfersiorfik area in the northern CNO is intruded by quartz–diorite and tonalite of the Arfersiorfik association and forms part of the allochthonous Ussuit unit (Manatschal et al., in press; van Gool et al., this issue). These rocks are primarily layered pelitic to psammitic metasedimentary rocks and banded amphibolites, with subordinate thin marbles and calc–silicate rocks. The latter commonly occur at the tectonic contacts of the Ussuit unit with the Archaean gneisses together with discontinuous units of banded amphibolite (deformed meta-pillow basalt?) and lenses of metaperidotite. In a few locations, thin (generally ≤ 5 m thick) quartzites and quartzo-feldspathic metasediments occur below the Ussuit unit and show no high-strain contact with the Archaean gneisses below. These appear not to be part of the allochthonous Ussuit unit and are interpreted to represent an original cover sequence, depositional on the Archaean gneisses structurally below the main décollement.

The supracrustal rocks at the SNO–CNO boundary occur in a 10–15 km wide zone in which thin, but laterally continuous, steeply north-dipping panels of supracrustal rocks are interleaved with Archaean gneisses by thrust imbrication and are isoclinally folded on a kilometre-scale (van Gool et al., 1996; Marker et al., 1997b). These panels are dominated by anatectic pelitic to quartzitic metasediments, with subordinate marble and mafic lithologies, and are transitional from amphibolite to granulite facies. They have not been seen to be cut by the Kangâmiut dykes. These rocks have been deformed in the high strain zone at this tectonic boundary and the original relations with the Archaean gneisses are obscured.

The ages and origins of the metasediments in the three settings described above are reported in the present study, while other supracrustal suites in the orogen are presently under investigation (Marker et al., 1997a).

4. SHRIMP zircon geochronology

4.1. Analytical technique and data assessment

U–Th–Pb concentrations and isotopic ratios in zircons were determined using SHRIMP I and were referenced to the Australian National University standard zircon SL13 (572 Ma; $^{206}Pb/^{238}U = 0.0928$). Descriptions of analytical procedure and data assessment are given by Compston et al. (1984) and Claoué-Long et al. (1995). Comparative isotope dilution and SHRIMP analyses of zircons from several well-preserved Proterozoic and Archaean samples (Roddick and van Breemen, 1994; Ireland, 1995) demonstrate that SHRIMP $^{207}Pb/^{206}Pb$ ratios are accurate. As cross-check on the accuracy of $^{207}Pb/^{206}Pb$ ratios obtained with SHRIMP in this study, zircons from Mesoproterozoic anorthosite AS3 were analysed interspersed with some of the unknowns of metasedimentary rock 407307. Isotope dilution thermal ionisation analysis of AS3 zircons yielded concordant ages with a weighted mean $^{207}Pb/^{206}Pb$ age of 1099.1 ± 0.5 Ma (Paces and Miller, 1993). Four AS3 zircons run interspersed with sample 407307 yielded a weighted mean $^{207}Pb/^{206}Pb$ age of 1113 ± 26 Ma.

As the zircons in the Nagssugtoqidian orogen may have suffered ancient Pb loss, it is an advantage to make as many analyses as possible in the available machine time, in order to find as many near concordant sites as possible. This was achieved by four scans (some metasedimentary rocks) and five scans (some metasedimentary rocks and metaigneous rocks) with counting times of 40 s or 50 s on ^{206}Pb and 120 s to 150 s on ^{207}Pb on individual analyses. Analytical precision was not compromised by this, because a large spot (≥ 50 μm, where the grains allowed it) was used for analysis, increasing by a factor of 2 to 3 the yield of secondary ions obtained, relative to that from a typical (30 μm) spot.

Zircon ages for the dyke samples presented here are weighted means (2σ) derived from $^{207}Pb/^{206}Pb$ ratios of analyses from the isotopically least-disturbed sites (those having closest to concordant U/Pb ages and low non-radiogenic components of Pb), which from optical microscopy and in some cases cathodoluminescence imaging were ascertained to belong to single structural/age components of the zircons.

The decay constants and present-day $^{238}U/^{235}U$ value given by Steiger and Jäger (1977) were used to calculate the ages. Samples were collected during expeditions supported by the Danish Lithosphere Centre and the Geological Survey of Greenland (GGU, now incorporated in the Geological Survey of Denmark and Greenland). The six digit sample numbers refer to the files of the GGU. The results of the analyses are presented in Table 1. Results for samples 158078 and 413840 were previously presented without details by Kalsbeek and Nutman (1996) (data repository).

4.2. Kangâmiut dykes 158078 and 413794

Dyke sample 158078 comes from a small island between the village of Kangaamiut and the mouth of Søndre Strømfjord (Fig. 2). The field setting and Rb–Sr whole rock isotopic data on this dyke were reported respectively by Fahrig and Bridgwater (1976) and Kalsbeek et al. (1978). Sample 413794 comes from a small island on the north side of the mouth of Itilleq fjord (Fig. 2). In both cases, leucodioritic dyke centres, interpreted to be comagmatic with the main dolerite, were sampled. Both dykes yielded anhedral to prismatic, medium to large zircons (up to 200 μm long). Amongst the anhedral varieties, jagged and blade-like forms are particularly common (Fig. 3). Some grains contain elongate mineral inclusions (Fig. 3). The prismatic habit of some grains suggests that their growth was unimpeded by the presence of other crystals, whereas it appears that for the anhedral ones their growth in some directions was restricted by other crystals. This suggests that the zircons grew in a partially crystallised magma. All grains are colourless to pale yellow, non-metamict, and devoid of inherited older cores and younger metamorphic overgrowths. All analysed sites yielded rather high U contents and Th/U ratios, typically >0.75 (Table 1). In both samples most analyses were close to concordant with $^{207}Pb/^{206}Pb$ ages between ~ 2070 and 1940 Ma (Fig. 4). There is a slight tendency for those with younger $^{207}Pb/^{206}Pb$ ages to be more discordant.

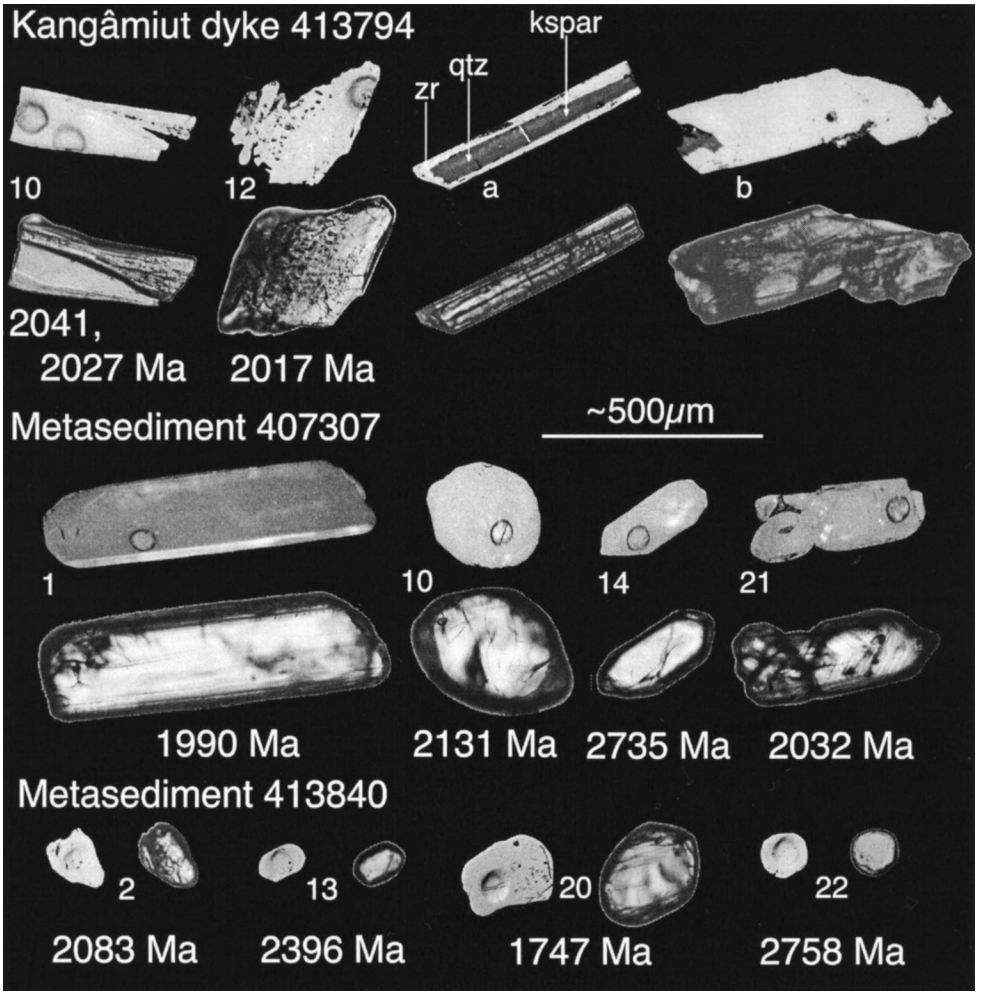


Fig. 3. Photomicrographs in transmitted and reflected light of zircons from Kangâmiut dyke 413794 and metasedimentary rocks 407307 and 413840. Grain numbers (Table 1) and $^{207}Pb/^{206}Pb$ apparent ages for analytical sites are given for each grain. Grains 'a' and 'b' for sample 413794 were not analysed, but have been included in order to show the morphological diversity of the grains in this sample. The elongate inclusion in grain 'a' consists of quartz (qtz) and K-feldspar (kspar). (Nutman, unpublished electron microprobe analyses).

This pattern of $^{207}Pb/^{206}Pb$ ages and the structural simplicity of the analysed sites suggests a magmatic population in excess of 2000 Ma with Pb loss during one or more younger events. With this assumption, rejection of analyses with the youngest $^{207}Pb/^{206}Pb$ ages until the remainder are indistinguishable from their common weighted mean, yielded indistinguishable ages of 2036 ± 5 Ma (2σ, $n=10$) for 413794 and 2046 ± 8 Ma (2σ, $n=9$) for 158078.

4.3. Quartzo-feldspathic metasedimentary rock 407307 and pelitic metasediment 413728, eastern CNO

Sample 407307 comes from an extensive sequence of metasediments of the Ussuit unit, intruded by quartz diorite and tonalite of the Arfersiorfik association, near the Inland Ice in the Nordre Strømfjord area (Fig. 2). It yielded abundant medium to large zircons (some >250 μm

Table 1
SHRIMP U–Pb zircon analyses

Spot	U (ppm)	Th/U	f206%	$^{206}Pb/^{238}U$ ratio	$^{207}Pb/^{206}Pb$ ratio	$^{207}Pb/^{206}Pb$ age	%disc
158078, Kangâmiut dyke, north of Kangaamiut village							
1-1	495	1.20	0.04	0.366±4	0.1262±12	2046±17	−2
2-1	750	1.05	0.10	0.358±4	0.1190±10	1942±15	2
3-1	748	0.99	0.01	0.373±4	0.1258±5	2039±7	0
4-1	393	0.49	0.05	0.361±4	0.1268±8	2054±12	−3
5-1	629	0.63	0.09	0.353±4	0.1245±6	2022±8	−4
6-1	503	0.76	0.01	0.375±4	0.1256±6	2037±9	1
7-1	216	0.75	0.22	0.331±4	0.1220±14	1986±21	−7
8-1	336	0.65	0.02	0.372±5	0.1280±16	2071±22	−2
9-1	398	0.81	0.21	0.341±5	0.1245±11	2022±16	−6
10-1	182	0.61	0.21	0.365±7	0.1272±12	2060±16	−3
11-1	427	1.02	0.08	0.357±5	0.1276±7	2065±10	−5
11-2	509	0.89	0.21	0.353±5	0.1264±7	2048±10	−5
413794, Kangâmiut dyke, near mouth of Itilleq (fjord)							
1-1	3482	1.07	0.02	0.368±7	0.1261±5	2045±7	−1
2-1	907	1.01	0.01	0.361±7	0.1257±10	2039±14	−3
3-1	1210	1.15	0.04	0.338±7	0.1225±9	1993±13	−6
4-1	2635	0.83	0.01	0.365±7	0.1255±3	2036±5	−1
5-1	1098	1.31	0.02	0.347±7	0.1252±5	2031±7	−5
6-1	1092	1.12	0.02	0.336±7	0.1221±15	1988±23	−6
6-2	1501	1.13	0.02	0.353±7	0.1233±4	2005±6	−3
7-1	578	0.91	0.06	0.353±8	0.1245±15	2022±22	−4
8-1	896	1.00	0.02	0.366±8	0.1246±8	2024±12	−1
9-1	1829	1.33	0.03	0.361±10	0.1259±5	2041±7	−3
10-1	1083	1.13	0.03	0.362±7	0.1260±3	2041±4	−2
10-2	959	0.82	0.07	0.319±6	0.1249±3	2027±5	−12
11-1	1138	1.01	0.04	0.342±7	0.1218±6	1983±9	−4
12-1	1298	1.53	0.04	0.329±6	0.1242±4	2017±6	−9
13-1	546	0.91	0.05	0.337±8	0.1249±15	2028±21	−8
407307, quartzo-feldspathic sediment, eastern CNO							
1-1	1178	0.64	0.03	0.350±6	0.1223±11	1990±15	−3
2-1	227	0.41	0.02	0.362±6	0.1226±19	1994±28	0
3-1	565	0.03	0.02	0.362±6	0.1244±6	2020±9	−2
4-1	2236	0.42	0.02	0.367±6	0.1279±3	2070±5	−3
5-1	150	0.44	<0.01	0.401±6	0.1388±14	2212±18	−2
6-1	282	0.47	0.25	0.377±6	0.1278±11	2068±15	0
7-1	514	0.13	0.36	0.354±6	0.1233±8	2004±11	−3
8-1	1050	0.36	0.04	0.371±24	0.1241±173	2016±270	+1
9-1	223	0.55	0.77	0.375±7	0.1242±18	2018±26	+2
10-1	127	0.45	1.13	0.388±7	0.1325±30	2131±40	−1
11-1	582	0.50	0.16	0.377±6	0.1265±7	2049±10	+1
12-1	144	0.92	1.07	0.374±8	0.1228±26	1997±39	+3
13-1	268	0.60	0.05	0.395±6	0.1373±13	2193±16	−2
14-1	534	0.06	0.07	0.483±7	0.1892±14	2735±12	−7
15-1	248	0.87	<0.01	0.376±8	0.1248±6	2026±9	+2
16-1	304	0.60	0.33	0.393±6	0.1258±23	2041±32	+3
17-1	319	0.68	0.16	0.363±6	0.1254±10	2035±14	−2
18-1	734	0.48	0.16	0.365±6	0.1257±6	2039±9	−2
19-1	115	0.47	0.02	0.371±6	0.1336±34	2146±45	−5
20-1	1300	0.22	0.01	0.337±5	0.1196±6	1950±9	−4
21-1	1292	0.26	0.02	0.353±5	0.1246±6	2032±9	−4

Table 1 (*continued*)
SHRIMP U–Pb zircon analyses

Spot	U (ppm)	Th/U	f206%	$^{206}Pb/^{238}U$ ratio	$^{207}Pb/^{206}Pb$ ratio	$^{207}Pb/^{206}Pb$ age	%disc
22-1	160	0.65	<0.01	0.376±6	0.1316±9	2119±12	−3
23-1	351	0.15	0.29	0.360±5	0.1245±13	2022±19	−2
24-1	1231	0.19	<0.01	0.332±5	0.1235±3	2008±5	−8
25-1	1318	0.33	0.06	0.349±48	0.1194±28	1947±42	−1
26-1	1294	0.39	0.01	0.351±5	0.1245±6	2022±8	−4
27-1	147	0.41	0.07	0.434±8	0.1581±22	2435±24	−5
28-1	642	0.10	0.03	0.374±6	0.1336±5	2146±7	−5
29-1	672	0.18	0.41	0.297±4	0.1150±8	1881±12	−11
29-2	378	0.37	0.37	0.336±5	0.1248±13	2026±19	−8
30-1	164	0.48	0.73	0.355±6	0.1184±24	1932±37	+1
31-1	187	0.28	0.64	0.365±6	0.1229±20	1998±29	0
32-1	118	0.69	0.80	0.377±16	0.1241±40	2016±63	+2
33-1	206	1.95	0.32	0.349±7	0.1203±29	1960±43	−2
34-1	392	0.29	0.32	0.347±6	0.1218±19	1983±27	−3
35-1	1129	0.12	0.10	0.361±5	0.1317±6	2120±9	−6
36-1	450	0.47	0.11	0.389±6	0.1361±9	2179±12	−3
37-1	761	0.37	0.04	0.344±5	0.1238±8	2011±11	−5
38-1	683	0.33	0.03	0.347±12	0.1225±12	1992±17	−4
39-1	1131	1.54	0.10	0.388±6	0.1407±5	2235±6	−5
40-1	138	0.95	0.27	0.561±9	0.2184±16	2969±12	−3
413728 mica schist, eastern CNO							
1-1	1977	0.09	0.03	0.355±7	0.1286±3	2079±5	−6
2-1	455	0.37	0.14	0.329±7	0.1253±7	2032±10	−10
3-1	334	0.56	0.07	0.353±11	0.1256±9	2038±13	−4
4-1	326	0.88	1.10	0.398±11	0.1455±13	2293±15	−6
5-1	376	0.26	0.15	0.336±9	0.1193±9	1945±14	−4
6-1	238	0.32	0.14	0.339±10	0.1246±39	2023±55	−7
7-1	202	1.65	0.24	0.419±14	0.1203±9	1961±13	15
8-1	249	0.38	0.12	0.519±22	0.1808±55	2661±50	1
9-1	494	0.57	0.55	0.338±7	0.1225±8	1993±12	−6
10-1	97	0.26	0.64	0.335±9	0.1175±19	1919±28	−3
11-1	223	0.29	0.32	0.364±10	0.1243±27	2019±39	−1
12-1	438	0.77	0.04	0.368±10	0.1256±8	2037±11	−1
13-1	186	0.33	0.66	0.346±9	0.1234±19	2004±28	−4
14-1	191	0.51	0.21	0.355±8	0.1258±11	2040±15	−4
15-1	2320	0.56	1.37	0.318±8	0.1218±23	1983±33	−10
16-1	294	0.30	0.35	0.315±8	0.1199±28	1955±41	−10
17-1*	268	0.34	0.13	0.317±7	0.1129±9	1847±14	−4
17-2*	435	0.60	0.07	0.309±8	0.1145±12	1872±17	−7
18-1	449	0.83	0.11	0.336±7	0.1227±6	1996±8	−7
19-1	293	0.66	0.37	0.351±8	0.1229±16	1999±23	−3
20-1	593	0.51	0.04	0.344±8	0.1250±6	2029±7	−6
21-1	480	0.43	0.08	0.339±8	0.1227±26	1996±38	−6
22-1	350	0.56	0.04	0.379±16	0.1487±35	2331±40	−11
23-1	441	0.77	0.16	0.345±8	0.1252±9	2031±13	−6
24-1	637	0.21	0.08	0.356±8	0.1238±6	2012±8	−3
25-1	178	0.36	0.42	0.372±9	0.1326±12	2133±16	−4
26-1	396	0.30	0.01	0.339±8	0.1217±12	1981±17	−5
27-1	863	0.30	0.05	0.330±7	0.1231±6	2002±9	−8
28-1	508	0.35	0.15	0.380±9	0.1385±6	2209±8	−6
29-1	119	0.55	0.52	0.339±12	0.1235±38	2007±55	−6
30-1	432	0.25	0.72	0.310±7	0.1209±12	1970±18	−12

Table 1 (*continued*)
SHRIMP U–Pb zircon analyses

Spot	U (ppm)	Th/U	f206%	$^{206}Pb/^{238}U$ ratio	$^{207}Pb/^{206}Pb$ ratio	$^{207}Pb/^{206}Pb$ age	%disc
429380 quartzitic paragneiss, eastern CNO							
1-1	30	0.60	1.22	0.575±23	0.2510±68	3191±43	−8
2-1	64	1.00	0.02	0.674±22	0.2821±15	3375±8	−2
3-1	176	0.27	1.73	0.357±8	0.1416±21	2247±26	−12
4-1	140	0.52	0.14	0.582±12	0.2184±15	2967±11	0
5-1	353	0.45	0.25	0.494±13	0.1830±26	2680±23	−3
6-1	323	0.77	0.03	0.646±15	0.2700±12	3306±7	−3
7-1	166	0.63	0.27	0.514±12	0.1903±31	2745±27	−3
8-1	437	0.58	0.03	0.377±10	0.1339±9	2150±11	−4
9-1	94	0.22	0.67	0.596±24	0.2291±66	3046±47	−1
10-1	205	0.48	0.46	0.363±9	0.1732±15	2588±14	−23
11-1	87	0.67	0.49	0.552±19	0.2467±55	3163±36	−10
12-1	240	0.24	0.06	0.496±15	0.1909±49	2750±43	−6
13-1	478	0.91	0.05	0.600±14	0.2826±39	3377±22	−10
14-1	187	1.03	0.19	0.658±21	0.2789±15	3357±8	−3
15-1	258	0.28	0.26	0.490±15	0.1865±45	2712±40	−5
16-1	292	0.89	0.29	0.466±11	0.1845±11	2693±10	−8
413840, mica schist, boundary between SNO and CNO							
1-1	156	0.69	0.23	0.465±13	0.1841±38	2691±35	−9
2-1	419	0.10	0.09	0.364±9	0.1289±18	2083±25	−4
3-1	273	0.02	0.14	0.317±8	0.1149±18	1878±29	−5
4-1	206	0.56	<0.01	0.403±12	0.1545±18	2396±20	−9
5-1	126	0.30	0.20	0.401±14	0.1521±30	2370±34	−8
6-1	228	0.33	0.19	0.416±17	0.1559±8	2412±89	−7
7-1	78	0.32	0.78	0.337±10	0.1090±24	1783±40	+5
8−1	407	0.12	0.11	0.338±8	0.1130±7	1847±11	+1
9-1	239	0.09	0.19	0.384±10	0.1325±13	2132±18	−2
10-1	313	0.29	0.39	0.399±11	0.1525±36	2374±40	−9
11-1	79	0.94	0.45	0.561±29	0.1976±68	2806±57	+2
12-1	386	0.27	0.19	0.429±13	0.1558±39	2411±43	−4
13-1	422	0.19	0.29	0.420±10	0.1545±10	2396±11	−6
14-1	139	0.05	0.41	0.332±12	0.1116±28	1826±45	+1
15-1	274	0.09	0.48	0.336±8	0.1174±11	1918±17	−3
16-1	170	0.06	0.22	0.312±10	0.1107±15	1810±24	−3
17-1	253	0.29	0.20	0.486±13	0.1823±12	2674±11	−5
18-1	259	0.42	0.12	0.438±20	0.1651±37	2509±38	−7
19-1	289	0.45	0.24	0.399±11	0.1471±16	2312±19	−6
20-1	115	0.09	0.81	0.316±11	0.1069±28	1747±49	+1
21−1	92	0.74	0.11	0.507±27	0.1955±84	2789±72	−5
22-1	189	0.21	0.06	0.510±16	0.1919±52	2758±45	−4
23-1	215	0.43	0.12	0.404±12	0.1528±20	2377±22	−8
24-1	245	0.66	0.12	0.467±12	0.1774±11	2628±10	−6
25-1	434	0.27	0.13	0.426±10	0.1671±8	2529±8	−9
26-1	261	0.20	0.03	0.433±16	0.1577±42	2432±45	−5
27-1	420	0.08	0.13	0.360±10	0.1305±19	2104±25	−6
28-1	236	0.13	0.22	0.459±13	0.1556±32	2409±35	+1
29-1	358	0.41	0.25	0.378±11	0.1424±9	2256±11	−8
30-1	214	0.43	0.15	0.518±16	0.2038±16	2857±13	−6

f206% is the proportion of ^{206}Pb in per cent that is not radiogenic.
%disc is the degree of discordance in per cent between the $^{207}Pb/^{206}Pb$ and $^{206}Pb/^{238}U$ ages.
All errors are quoted at the 1σ level.

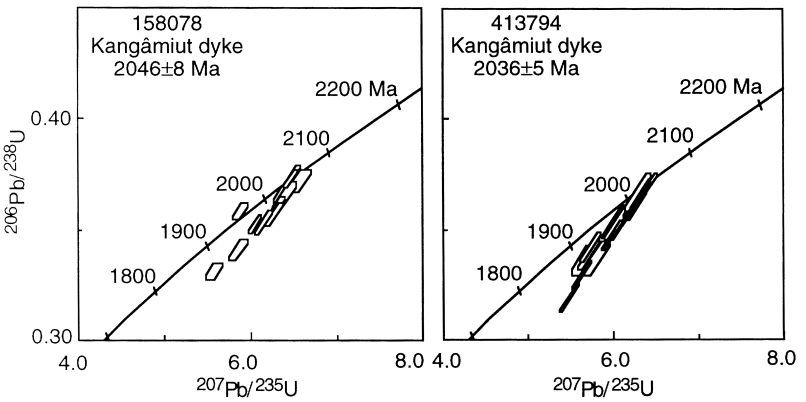


Fig. 4. U–Pb concordia diagrams for zircons from Kangâmiut dykes 158078 and 413794. Errors are depicted at the 1σ level.

long). The grains are mostly prismatic with a lesser number ovoid in shape. The pyramidal terminations of all grains have been rounded or knocked off (Fig. 3). Most grains show well-developed micron-scale concentric zoning, which is commonly truncated at the rounded grain boundaries. However, a few structureless grains are also present, and some grains are partly metamict. A few grains show small protuberances, which might be outgrowths of new metamorphic zircon, but these were too small to be analysed. The zircons are interpreted to be detrital grains, with rare in situ growth of new metamorphic zircon. Many of the zircons are rich in U (Table 1) and nearly all grains yielded close to concordant ages, mostly with $^{207}Pb/^{206}Pb$ ages of 1950–2100 Ma, with a lesser number of ~2200 Ma and only a few Archaean grains (Fig. 5).

The age of deposition is constrained by the age of the youngest detrital zircons (~1950 Ma) and the intrusion of rocks of the Arfersiorfik association (~1920 Ma). The predominance of 1950–2100 Ma zircons suggests that the sediment was derived mostly from a Palaeoproterozoic terrane, with only a very minor direct contribution from older crust. The lack of considerable rounding of the detrital grains and their large size suggests they may have experienced only limited transport in the sedimentary system and thus had a proximal source.

Pelitic metasediment 413728 is from the coast of inner Nordre Strømfjord (Fig. 2). It is from a strongly deformed, migmatised, garnetiferous paragneiss, which is part of the Ussuit unit, sandwiched between quartz–diorite of the Arfersiorfik association and a seam of marble, which separates it from the Archaean gneisses below. The sample yielded abundant brown to yellow zircons, mostly 100 to 150 μm long. Most grains are prismatic, but some ovoid grains occur. The terminations of the prismatic grains are rounded, similar to those in sample 407307. Some grains show embayments on their prismatic faces, and a few very small protuberances, possibly new metamorphic zircon growth, are also present. Grain 17 (Table 1) is prismatic, but has lost one pyramidal termination, possibly during mineral separation. The other termination is sharp and euhedral, in contrast to all other grains in the sample. It might contain a small inherited core. Many of the grains have high U concentrations, but not as extreme as in sample 407307 (Table 1). The sample yielded mostly 1950–2100 Ma grains, with a few older grains, comparable with the population in sample 407307 (Table 1; Fig. 5). Duplicate analyses were undertaken on euhedral grain 17. Analysis 17-1 of a completely clear part of the grain, yielded a $^{207}Pb/^{206}Pb$ age of 1847±28 Ma (2σ), whereas a slightly cloudy area of the grain yielded a $^{207}Pb/^{206}Pb$ age of 1872±34 Ma (2σ), and was slightly more discordant.

The zircon data suggest a similar age of deposition and provenance as sample 407307. The results from grain 17 are an indication of new zircon

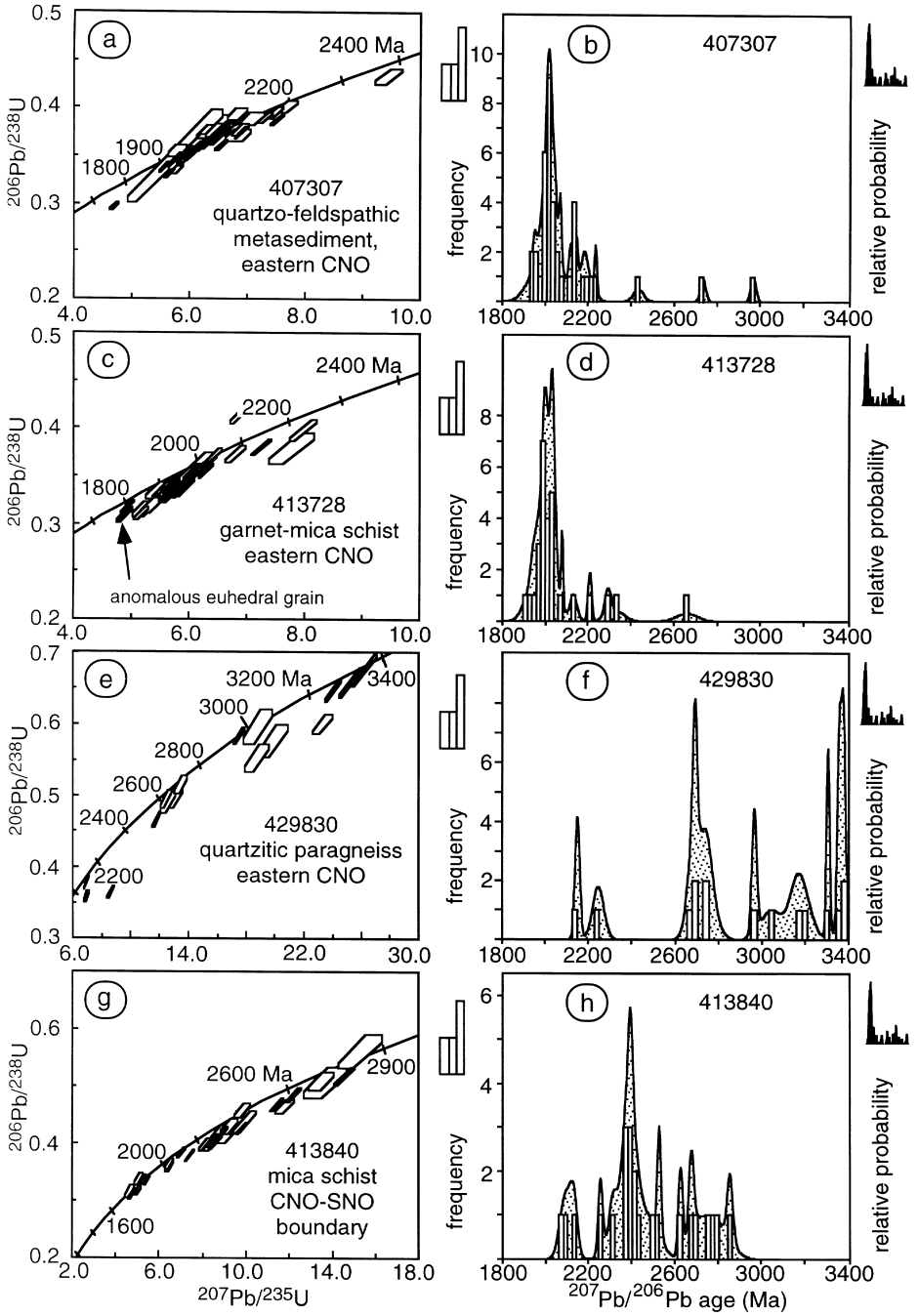


Fig. 5. U–Pb concordia diagrams, $^{207}Pb/^{206}Pb$ age histograms and (dimensionless) relative probability distribution plots for metasedimentary rocks 407307, 413728, 429830 and 413840. The U–Pb concordia diagrams show all data (errors depicted at the 1σ level), whereas the $^{207}Pb/^{206}Pb$ age plots incorporate only detrital grain analyses $<10\%$ discordant.

growth during a thermal event between 1800 and 1900 Ma, in agreement with the age of 1845 ± 23 Ma for the peak of metamorphism obtained by Taylor and Kalsbeek (1990).

4.4. Quartz-rich metasedimentary rock 429830, eastern CNO

Sample 429830 belongs to a unit of ortho- and paragneisses in the core of a large E-plunging antiform, in the southern arm of Nordre Strømfjord (Fig. 2). It is a quartz-rich metasedimentary rock from a package of diffusely foliated biotite- and cordierite-bearing rocks which is separated from the overlying metasedimentary rocks of the Ussuit unit by a tectonic contact, marked by lenses of ultramafic rocks and thin marbles.

The quartz-rich metasediment 429830 yielded small to medium ovoid and prismatic zircons (mostly ≤ 100 μm long). The pyramidal terminations of the prismatic grains have been rounded. Some grains display micron-scale concentric zoning, which in some cases is truncated at the grain boundaries. Other grains are colourless and show little or no concentric zoning. At least half of the grains have thin incomplete rinds, < 5 μm wide, of new zircon growth, pale yellow in colour. These rinds are so thin that they are beyond the spatial resolution of the SHRIMP. The zircons are interpreted to be detrital grains, affected by some in situ growth of new metamorphic zircon.

16 grains were analysed, all of which have a low abundance of non-radiogenic Pb (Table 1). Most grains yielded analyses $< 10\%$ discordant (Fig. 5), with $^{207}Pb/^{206}Pb$ ages between 2150 and 3375 Ma. Rather than there being a continuum of $^{207}Pb/^{206}Pb$ ages, there are groups at 3300–3400 Ma (4), ~ 3200 Ma (2), ~ 3000 Ma (2), 2650–2750 Ma (5) and 2150–2250 Ma (2). The youngest grains (3 and 8) are clear and ovoid, and there are no analytical or petrographic reasons to regard them as metamorphic in origin or strongly disturbed Archaean grains. Therefore, metasedimentary rock 429830 is interpreted as deposited at < 2150 Ma, but derived largely from Archaean sources.

4.5. Metasedimentary rock 413840, CNO–SNO boundary

The metasedimentary rock 413840 is from a thin unit of slightly weathered sillimanite + kyanite schist interlayered with thicker quartzites and associated with marble horizons, which together form one of the panels of supracrustal rocks tectonically interleaved with Archaean gneisses along the CNO–SNO boundary (Fig. 2). The sample yielded small ovoid and less common small rounded prismatic grains (many only ~ 50 μm across). These grains are interpreted to be detrital in origin, their small size consistent with the likely pelitic protolith of the sample. Some of these grains have more euhedral protuberances and outgrowths of new zircon, in some cases as large as the original grains. Some larger (~ 100 μm) multifaceted equant grains are also present (Fig. 3).

Analyses of larger equant grains and overgrowths show moderate to low U contents, and variable, but commonly very low Th/U values (Table 1). They yielded a weighted mean $^{207}Pb/^{206}Pb$ age of 1840 ± 19 Ma (2σ, $n = 5$; after rejecting one with a low $^{207}Pb/^{206}Pb$ age), interpreted as giving the time of metamorphism of the sediment. The small ovoid and rounded prismatic grains yielded close to concordant ages. There is a tendency for all analyses to be a few percent discordant, suggesting minor recent Pb loss, perhaps during recent weathering (Table 1; Fig. 5). $^{207}Pb/^{206}Pb$ ages for these grains range from ~ 2100 Ma to ~ 2850 Ma, with a majority of grains with ages around 2400 Ma. It is possible that all these grains are Archaean in age, and that many lost a high proportion of their radiogenic Pb in the ~ 1840 Ma event. However, such Pb loss is unlikely to give rise to the clustering of ages at ~ 2400 Ma, in grains of different sizes and U contents. Therefore we consider it more likely that the detrital population is dominated by ~ 2400 Ma grains, with lesser amounts of younger (two ~ 2100 Ma) and older (Archaean) grains, variably affected by minor loss of radiogenic Pb. These results indicate that the sediment was deposited in the Palaeoproterozoic, after ~ 2100 Ma, the youngest detrital grains, but before 1840 ± 19 Ma, the age of metamorphism. The absence of distinct

Kangâmiut dykes within the metasedimentary rocks suggests deposition after ~2040 Ma. The main sources of detrital material were considerably older.

5. Discussion

5.1. The Kangâmiut dyke swarm

After stabilisation of the Archaean North Atlantic craton at the end of the Archaean, tectonic activity in this continental block was characterised by the intrusion of a number of mafic dyke swarms, which continued from ~2.5 to 1.9 Ga and was followed by (partially overlapping) Palaeoproterozoic orogenesis in the surrounding orogenic belts (Bridgwater et al., 1995 and references cited therein). Dyke activity was concentrated in swarms that were restricted in time and space, although partially overlapping, each marking a specific thermal and tectonic regime (Bridgwater et al., 1995).

Two dated Kangâmiut dykes from ~50 km apart yielded ages of ~2040 Ma and represent one of the youngest phases of widespread Palaeoproterozoic mafic dyke activity in the North Atlantic craton. A less precise but indistinguishable age of 2048 ± 17 Ma has been obtained on a dyke from within the continuation of the Nagssugtoqidian belt in East Greenland (Nutman and Bridgwater, unpublished data). This suggests not only that the Kangâmiut dykes form a swarm that stretches beneath the ice-cap from the west coast to the east coast of Greenland, and forms one of the larger swarms in the Archaean craton, but also that the dykes are spatially related to the Nagssugtoqidian orogen. However, the age of 2040 Ma indicates that the emplacement of these dykes and the tectonic regime operating at that time predates the intrusion of all known calc alkaline units (Arfersiorfik association and Sisimiut charnockite suite) by ~100 Ma and subsequent collisional events (during ~1850–1800 Ma metamorphism) in the Nagssugtoqidian orogen by ~200 Ma.

The tectonic significance of the Kangâmiut dykes and their genetic relation to the orogen are yet poorly understood, but work is in progress to look at the conditions and tectonic setting of dyke emplacement. The synkinematic nature of the dykes indicates that the area was tectonically active 100 Ma before the onset of orogenesis. The suggestion that they were emplaced in a transpressive regime (Escher et al., 1976b; Hanmer et al., 1997) makes their interpretation as simple rift magmatism associated with opening of an oceanic basin unlikely.

5.2. Correlation of metasedimentary rocks in the Ussuit unit

The age of deposition and the dominance of 1950–2100 Ma detrital zircons in the analysed sediments of the Ussuit unit (samples 407307 and 413728), is similar to the detrital zircon populations in samples of metasedimentary rocks in the Torngat orogen in northern Labrador and on Baffin Island (Tasiuyak gneisses; Scott and Machado, 1994; Scott and Gauthier, 1996) and in the Rinkian belt to the north of the Nagssugtoqidian orogen (Nûkavsak Formation; Kalsbeek et al., in press). Similar to the situation in the Nagssugtoqidian orogen, these detrital zircon populations of around 1950–2100 Ma have no known source within the Torngat and Rinkian belts respectively. The psammitic to pelitic Tasiuyak gneisses are interpreted to be deposited in an accretionary wedge (Van Kranendonk and Wardle, 1994), which was later emplaced as an allochthon onto the Nain continent (Rivers et al., 1996). The Nûkavsak Formation in the Rinkian belt is a flysch deposit forming part of the Karrat Group, in which it stratigraphically overlies the lower Marmorilik and Qeqertarssuaq formations (Henderson and Pulvertaft, 1987). The latter two formations consist of shelf sediments (quartzite, marble, pelitic schists), while the Nûkavsak Formation (metagreywackes and metapelites) might have been deposited in a foreland basin (Grocott and Pulvertaft, 1990). The highly disrupted nature of the sediments of the Ussuit unit prevents a direct correlation based on depositional environment, but the allochthonous nature of Tasiuyak gneisses and Ussuit metasediments is comparable.

5.3. Older detritus in Palaeoproterozoic metasediments

Metasedimentary rock 413840 is from a series of thrusted quartzo-feldspathic and pelitic metasedimentary rocks at the SNO–CNO boundary while 429830 represents a unit of quartzitic metasedimentary rocks tectonically beneath the allochthonous Ussuit unit. Although the time of deposition may have overlapped with that of the sediments in the Ussuit unit, they were derived from different sources. The dominance of Archaean zircons in 429830, and $\geq$2400 Ma zircons in 413840, together with a few 2100–2200 Ma grains, and the association with marble tectonites suggests that these metasedimentary rocks could have belonged to Palaeoproterozoic shelf sequences, comparable with the Marmorilik and Qeqertarssuaq formations in the Rinkian belt. Both 413840 and 429830 contain some $\sim$2700–2800 Ma detrital grains, in accord with known ages of Archaean rocks in the area (e.g. Kalsbeek and Nutman, 1996). 413840 from the CNO–SNO boundary contains a large population of $\sim$2400 Ma zircons. Likely source rocks of this age are unknown in Greenland, but are known from Kola and Karelia (Balashov et al., 1993; Bogdanova and Bibikova, 1993; Bogdanova et al., 1996; Zinger et al., 1996), areas that contain possible easterly continuations of the Nagssugtoqidian orogen (e.g. Bridgwater et al., 1990). The 3300–3400 Ma detrital zircons found in 429830 of the eastern CNO are the oldest recognised material in the Nagssugtoqidian orogen, and do not match the ages of any rocks in the immediate basement. The nearest match can be found in pegmatites in the Isua supracrustal belt (Baadsgaard et al., 1986), which, however, form only a tiny proportion of the Archaean craton. Thus it is likely that these Palaeoproterozoic metasedimentary rocks incorporate clastic material from a distal source. The predominance of $>$2400 Ma grains in them shows that they have a different source and origin from the allochthonous metasedimentary rocks 407307 and 413728 of the Ussuit unit, which are of approximately the same depositional age.

5.4. Palaeoproterozoic magmatic province

The 1950–2100 Ma ages of poorly rounded, large detrital zircons in the metasedimentary rocks 407307 and 413728 from the allochthonous Ussuit unit in the CNO suggest that these sediments were derived from nearby 1950–2100 Ma igneous rocks (arcs?). No granitoid rocks of 2100–1950 Ma have been found within the Nagssugtoqidian orogen. This could indicate that the allochthons carrying the sediments are far-traveled, or that source-rocks have been removed by erosion or are presently covered by the Inland Ice. The correlation of these sediments with those containing similar zircon populations in the Rinkian belt and the Torngat orogen, indicate that distribution of material of this age is widespread. Furthermore, the vast volumes of sedimentary material in the Tasiuyak gneisses (up to 13 km thick and exposed $>$500 km along strike, Rivers et al., 1996) and the Nûkavsak Formation (up to 4 km thick and exposed more than 400 km along the west coast of Greenland, Henderson and Pulvertaft, 1987) suggests that the source was also of considerable size. Exhumation of the source (arc?) rocks occurred rather shortly after emplacement, since the sediments in the Ussuit unit containing $\sim$1950 Ma detrital zircons, were deposited before $\sim$1920 Ma, the age of intrusion of rocks of the Arfersiorfik association. The only exposed candidate source rocks known within Greenland are dioritic–tonalitic–granodioritic rocks of 2000–1950 Ma in the gneiss complexes which occur within the Caledonian orogenic belt of Northeast Greenland and in Inglefield Land of Northwest Greenland (Kalsbeek et al., 1993; Nutman and Kalsbeek, 1994; Kalsbeek and Nutman, unpublished data).

The coincidence in age between the dominant detrital zircon population in metasedimentary rocks 407307 and 413728 of the Ussuit unit and the Kangâmiut dykes is intriguing. The zircon geochronological results suggest that in a period at about 2040 Ma the southern foreland of the future Nagssugtoqidian orogen underwent deformation and emplacement of the extensive Kangâmiut dyke swarm, which was at least in part simultaneous with development of 1950–2100 Ma felsic igneous complexes (arcs?). Although this

similarity in age could suggest a genetic link, in the absence of firm data on the environment and tectonic setting of the Kangâmiut dykes, a direct correlation would be highly speculative.

6. Conclusions

- Two mafic dykes in the Kangâmiut dyke swarm have ages of ~2.04 Ga.
- Two kinds of supracrustal suites were distinguished in the Nagssugtoqidian orogen. One, occurring in the allochthonous Ussuit unit, is dominated by detrital zircons of ~1950–2100 Ma; a second kind contains mainly much older zircon populations.
- Deposition of the allochthonous sediments took place between ~1950 and 1920 Ma; the others were deposited after ~2100 Ma and before ~1850 Ma.
- Metasediments in the first group were derived from ~1950–2100 Ma igneous rocks (arcs?) which overlap in age with the Kangâmiut dykes. A source of these zircons is not known within the Nagssugtoqidian orogen.
- Metasediments from the second group may represent remnants of original Palaeoproterozoic shelf sequences.

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